

## Numerical Analysis PhD Exam, January 2020

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Do 4 (four) of the first 5 (1–5) and 4 (four) of the last 5 problems (6–10).

1. Let  $A \in \mathbb{C}^{m \times n}$ .

- (a) Determine constants  $\alpha$  and  $\beta$  such that the following inequality holds for the  $p$  and  $\infty$  norms of matrix  $A$ , for integers  $p \geq 1$ . Justify your answer.

$$\alpha \|A\|_{\infty} \leq \|A\|_p \leq \beta \|A\|_{\infty}.$$

- (b) Prove or give a counterexample:  $\|A\|_2 \leq \|A\|_F$ , where  $\|A\|_F$  is the Frobenius norm of  $A$ . If you prove this, make sure to justify each nontrivial step.

2. Suppose  $A$  is Hermitian positive definite.

- (a) Prove that each principal submatrix of  $A$  is Hermitian positive definite.  
(b) Prove that an element of  $A$  with largest magnitude lies on the diagonal.  
(c) Prove that  $A$  has a Cholesky decomposition.

3. Define the matrices  $A$  and  $B$  by

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find both full and economy singular value decompositions of  $A$ .  
(b) Find both full and economy QR decompositions of  $B$ .

4. Let  $\|\cdot\|$  be a subordinate (induced) matrix norm.

- (a) If  $E$  is  $n \times n$  with  $\|E\| < 1$ , then show  $I + E$  is nonsingular and

$$\|(I + E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

- (b) If  $A$  is  $n \times n$  invertible and  $E$  is  $n \times n$  with  $\|A^{-1}\|\|E\| < 1$ , then show  $A + E$  is nonsingular and

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

5. Suppose the linear equation  $Ax = b$ , with

$$A = \begin{pmatrix} \delta & 1 \\ 1 & 1 \end{pmatrix}, \quad |\delta| < \epsilon_m/4,$$

is solved on a floating-point system with machine-epsilon  $\epsilon_m$ , using an LU factorization (no pivoting!) followed by forward and back substitution. (You may assume the operations  $(+, -, \div, \times)$ , do not incur any additional errors.)

- (a) If  $b = (1, 0)^T$ , compute the backward error,  $\|\hat{b} - b\|_2 / \|b\|_2$ , where  $\hat{b}$  is the data that satisfies  $A\hat{x} = \hat{b}$ , and  $\hat{x}$  is the computed solution.
- (b) Is the result of (a) sufficient to draw any conclusions about the backward-stability of the algorithm used to compute  $\hat{x}$ ? Explain.

6. This problem has two parts.

- (a) Starting with the first two Legendre polynomials over  $[-1, 1]$ , given by  $p_0(x) = 1$ , and  $p_1(x) = x$ , find the next three (monic) Legendre polynomials,  $p_2(x)$ ,  $p_3(x)$  and  $p_4(x)$ .
- (b) Find the nodes  $t_0, t_1, t_2$  and weights  $w_0, w_1, w_2$  which define the Gaussian Quadrature formula

$$\int_{-1}^1 f(t) dt \approx w_0 f(t_0) + w_1 f(t_1) + w_2 f(t_2),$$

with degree of exactness  $m = 5$ . You should find the nodes exactly, and may leave the weights  $w_0, w_1, w_2$ , in integral form.

7. Consider the fixed point problem  $x = f(x)$ , where  $f(x) = e^{-(3+x)}$ .

- (a) Assuming all computations are done in exact arithmetic, find the largest open interval in  $\mathbb{R}$  where the fixed-point iteration  $x_{k+1} = f(x_k)$  is ensured to converge.
- (b) Write a Newton iteration for finding the fixed-point.

8. Let  $P_1$  be the space of polynomials of degree at most one. Using the norm  $\|u\|_2 = \left(\int_a^b u^2 dx\right)^{1/2}$

- (a) Find the least-squares approximation to  $f(x) = x^3$  in  $P_1$  over  $[a, b] = [-1, 1]$ .
- (b) Find the least-squares approximation to  $f(x) = x^4$  in  $P_1$  over  $[a, b] = [0, 1]$ .

9. This problem has two parts.

- (a) Find a natural cubic spline  $B_0$  on  $-1 \leq x \leq 1$  that satisfies  $v(-1) = 0$ ,  $v(0) = 2$ , and  $v(1) = 0$ .
- (b) Find a natural cubic spline on  $-1 \leq x \leq 2$  that satisfies  $v(-1) = 0$ ,  $v(0) = 2$ ,  $v(1) = 1/2$  and  $v(2) = 0$ . You may write the solution as  $B = B_0 + B_1$ , where  $B_0$  is (or is closely related to) the solution to (a), if you like.

10. Let  $f \in C^\infty(a - H, a + H)$ , and let  $h < H$ . Let  $x_0 = a - h$ ,  $x_1 = a$  and  $x_2 = a + h$ .

- (a) Find the finite difference approximation to  $f''(a)$  based on the quadratic interpolant  $p_2$  which satisfies  $p_2(x_0) = f(x_0)$ ,  $p_2(x_1) = f(x_1)$  and  $p_2(x_2) = f(x_2)$  (you should explicitly show how the difference approximation is derived from the interpolant).
- (b) Let  $\psi_0(h) = \psi(h)$  be the difference approximation to  $f''(a)$  found in part (a). Assume (in exact arithmetic)  $\psi(h) \rightarrow \psi(0) = f''(a)$  as  $h \rightarrow 0$ , and that  $\psi(h)$  has the asymptotic expansion

$$\psi(h) = \psi(0) + a_2 h^2 + a_4 h^4 + a_6 h^6 + \dots$$

Find the general Richardson extrapolation formula for  $\psi_k(h)$  based on  $\psi_{k-1}(h)$  and  $\psi_{k-1}(h/2)$ .